

Parsa Oryani

M.Sc. Computer Engineering - Secure and Scalable Decentralized Systems

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Research Profile

M.Sc. student in Computer Engineering at Sharif University of Technology, specializing in Secure Computing. Research focus: security and scalability of decentralized systems, especially blockchain security and cross-chain/Layer-2 protocols.

Education

Sharif University of Technology

M.Sc. in Computer Engineering, Secure Computing

Sep 2025 - Present

Tehran, Iran

- National Master's Entrance Examination Rank: 56.
- Research focus: blockchain security, decentralized systems, cross-chain and Layer-2 protocols.
- Selected graduate coursework: Applied Cryptography (18.9/20), Secure Software Systems (18.2/20), Formal Methods in Information Security, Foundations and Applications of Blockchain, Deep Learning.

Amirkabir University of Technology (Tehran Polytechnic)

B.Sc. in Computer Science

Sep 2020 - Feb 2025

Tehran, Iran

- National University Entrance Examination Rank: 343.
- Strong preparation in cryptography, algorithms, probability, linear algebra, artificial intelligence, and software development.
- Cryptography coursework: Cryptography I (18.75/20); Special Topics in Cryptography (18.25/20, highest grade in class).

Research Experience

Research Assistant - Blockchain Security

Sharif University of Technology; Supervisor: Dr. Morteza Amini

Sep 2025 - Present

Tehran, Iran

- Conducting research on security and scalability challenges in blockchain and decentralized protocols, with emphasis on protocol-level security, interoperability, and blockchain infrastructure.
- Investigating cross-chain and Layer-2 systems, including state verification, trust assumptions, cross-rollup communication, and emerging attack surfaces.
- Analyzing protocol security assumptions, trust models, safety/liveness properties, and verification challenges in decentralized systems.
- Performing technical literature reviews and supporting implementation-oriented research with Ph.D. researcher Amirmohammad Aghapour.

Teaching Experience

Teaching Assistant, Computer Networks

Sharif University of Technology; Dr. Sadeghzadeh; undergraduate course

Feb 2026 - Jun 2026

Teaching Assistant, Machine Learning

Sharif University of Technology; Dr. Motahari; undergraduate course

Feb 2026 - Jun 2026

Teaching Assistant, Artificial Intelligence and Lab

Amirkabir University of Technology; Dr. Ghatee; undergraduate course

Feb 2025 - Jun 2025

Teaching Assistant, Foundations of Matrices and Linear Algebra

Amirkabir University of Technology; Dr. Najafi; undergraduate course

Sep 2023 - Jan 2024

Teaching Assistant, Advanced Programming

Amirkabir University of Technology; Dr. Bejani; undergraduate course

Feb 2023 - Jun 2023

Technical Research Toolkit

Protocol analysis	Security assumptions, trust models, interoperability risks, safety/liveness reasoning, and adversarial thinking for decentralized protocols.
Cryptographic systems	Applied cryptography, zero-knowledge proof concepts, Merkle structures, commitments, nullifiers, and privacy-preserving protocol design.
Blockchain stack	Ethereum/EVM, Hyperledger Besu, QBFT, JSON-RPC, transaction processing, mempool behavior, Layer-2 and cross-chain architectures.
Research engineering	Python, Rust, Solidity, TypeScript, C++, FastAPI, React, Docker, PostgreSQL, Linux, Git, reproducible experiment tooling.